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The Black Hole Equation: Explained

The Schwarzschild metric: an introduction

9 min read · Jun 25, 2022



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It's hard to express how utterly remarkable it is that mathematical scribbles can predict something as bizarre as black holes and then, looking to the skies with sufficient diligence, we find them.



A rotating black hole. [Source](#).

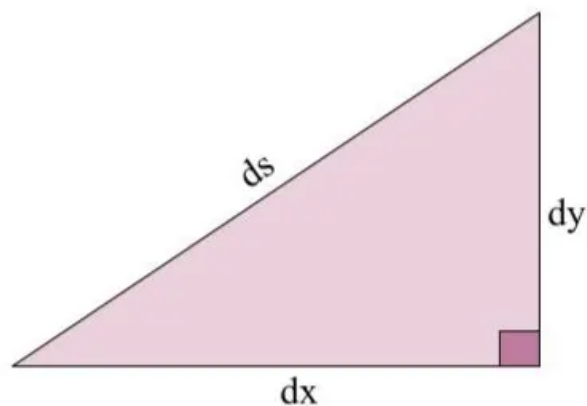
The equation which describes the behavior of space-time around stationary black holes is known as the Schwarzschild metric. It looks something like this:

$$\begin{aligned} ds^2 &= -c^2 d\tau^2 \\ &= -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \end{aligned}$$

The Schwarzschild metric

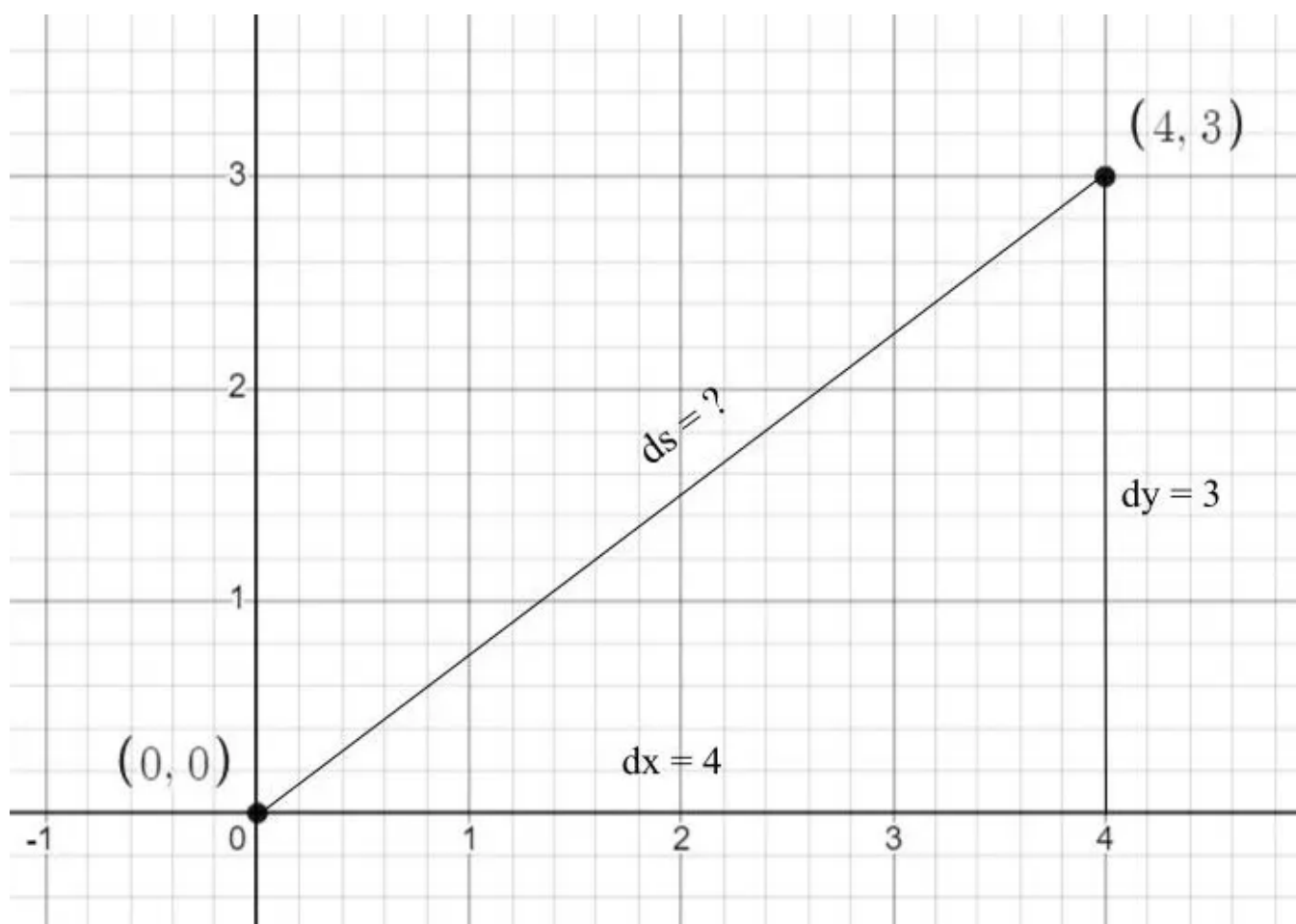
In Einstein's theory of general relativity, the Schwarzschild metric (also known as the Schwarzschild solution) is an exact solution to the Einstein field equations that describe the gravitational field outside a spherical mass, on the assumption that the electric charge of the mass, the angular momentum of the mass, and the universal cosmological constant are all zero. The solution is a useful approximation for describing slowly rotating astronomical objects such as many stars and planets, including Earth and the Sun. It was found by Karl Schwarzschild in 1916, and around the same time independently by Johannes Droste, who published his more complete and modern-looking discussion four months after Schwarzschild.

Let's kick this off with something seemingly unrelated: Pythagoras' theorem. I trust you're familiar with this.



$$ds^2 = dx^2 + dy^2$$

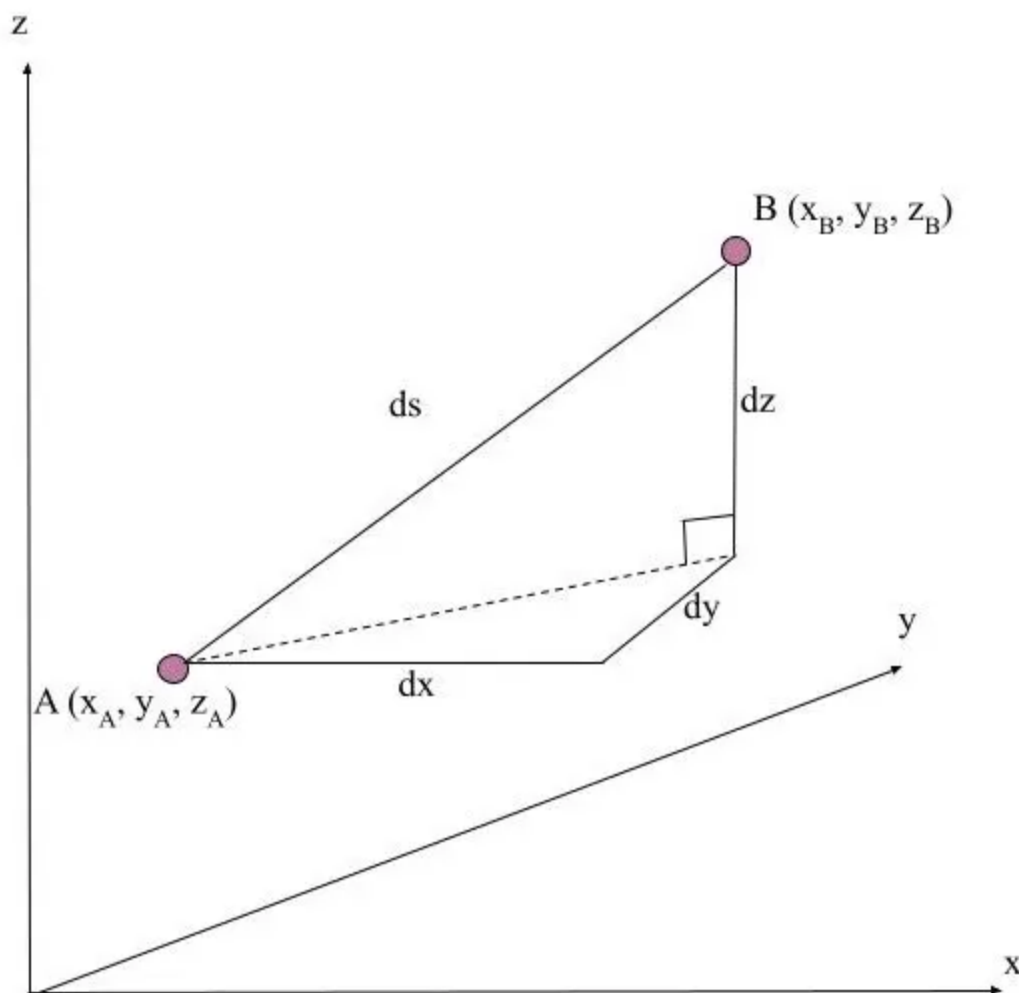
As long as we're given the length of two sides of a right-angled triangle, we can work out what the third one is. Now, this becomes particularly useful when we're trying to find the distance between two points on a coordinate grid.



To get from (0, 0) to (4, 3), we'd need to travel four units to the right and three units up. The distance between these two points will consequently be given by:

$$ds = \sqrt{dx^2 + dy^2}$$

With some straightforward math, we know that ds must be 5. Pythagoras' Theorem is incredibly useful when trying to find the distance between two points on a coordinate grid. But this can also be extended to 3-dimensional space.



Two points in 3D space.

Here, just $dy^2 + dx^2 = ds^2$ doesn't work too well. We haven't accounted for the z component. Naturally, our new equation becomes this:

$$ds = \sqrt{dx^2 + dy^2 + dz^2}$$

This equation gives us the distance between two points in a three-dimensional coordinate grid. The point I'm trying to make here is that when going from a two-dimensional grid to a three-dimensional one, we have to account for the extra spatial component. This works for the most part but special relativity is a theory

established in *four* dimensions, not three. There are three spatial dimensions followed by a time dimension. Our ds now looks something like this:

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$$

In other words, if we wanted to find the *interval* between two points in four-dimensional space, we'd need to account for the fourth dimension. This is exactly how we accounted for the third dimension when we went from the 2D grid to the 3D one. By the way, the reason why I called it the interval rather than the distance is because “distance” implies a spatial perspective. In this case, however, we’re considering an interval. A quantity that includes time.

Now, when we modify our formula and account for the time coordinate we include a c^2 term. This is because we move through time at the speed of light. So, our dt is scaled up by the speed of light. Relative to yourself, you do not move through space, so these velocities are zero. You then only move in the time-like direction, and in this direction, you move with the speed of light. Which, when we apply to Pythagoras’ theorem, we need to square.

The negative sign with the c^2 is irrelevant for now. It doesn’t really matter if we choose negative for the time and positive for the space or positive for the time and negative for the space. As long as we’re consistent, we’ll be met with the same answer.

Another caveat we want to consider when we invoke spacetime is that we’re not finding the interval between two points anymore; we’re finding the interval between two events. This is just embedded into the fundamental idea of spacetime. Not really important but that’s what we’ll call it nonetheless.

Anyway, we won’t be delving into the consequences of having a $-c^2$ term here. The idea remains the same regardless of its inclusion. It’s that we must account for an extra term when we consider higher dimensions. We haven’t even got to the black hole equation yet. But bear with me, we’re getting there. What’s important now is

we've calculated a framework for finding the interval between two events in four-dimensional space-time. However, the issue with this is that our equation assumes that space-time is flat. We've made an unsaid assumption here: the shortest distance between two points in space is a straight line. This is not always the case. It's why airplanes take curved paths. The shortest distance between two points in *curved* space is actually what we'd call a *geodesic*.

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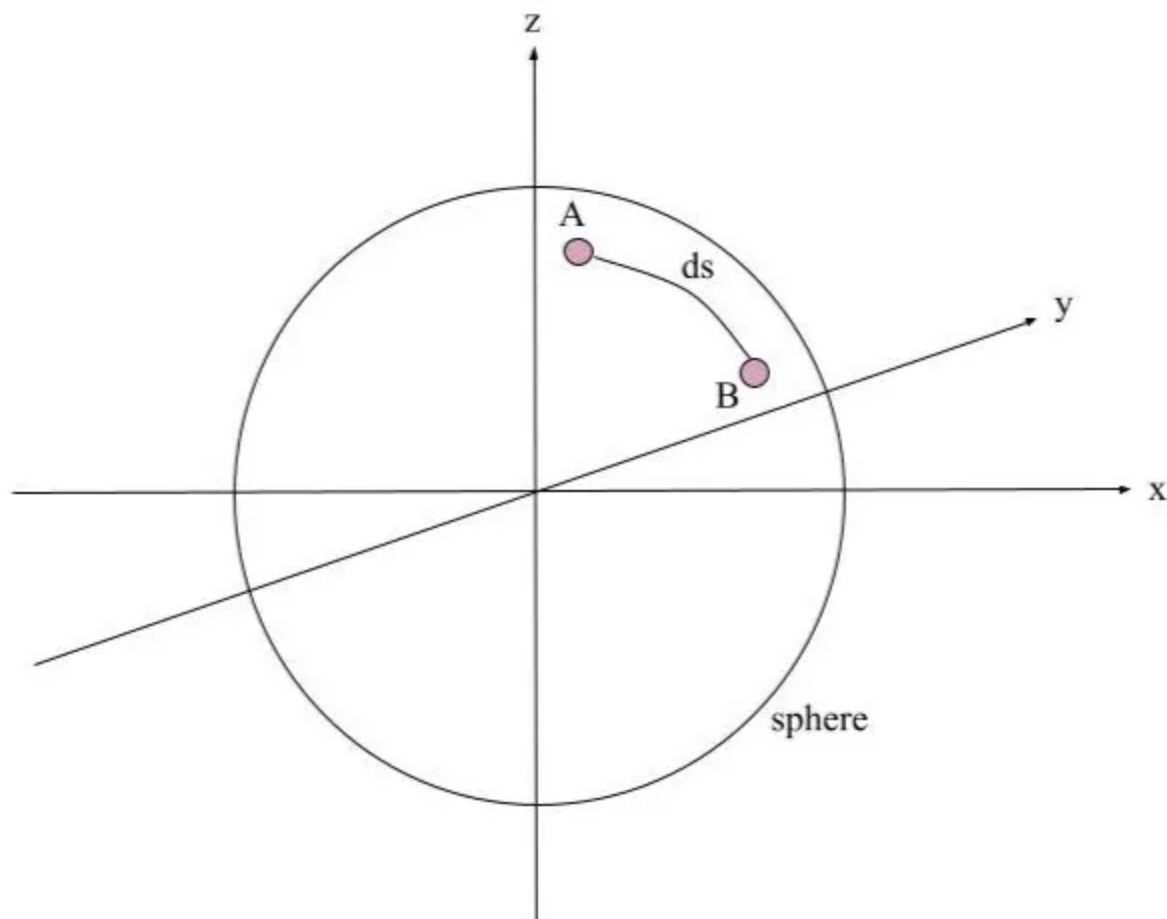
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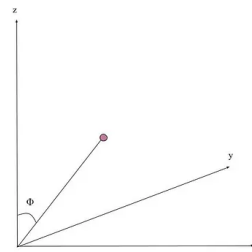
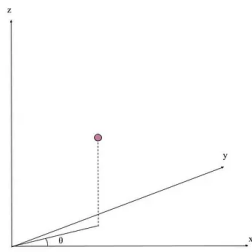
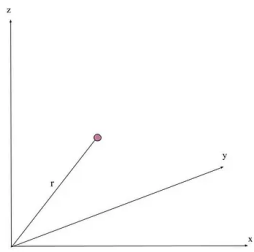
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This is usually difficult to imagine in four dimensions — partly because, well, we're terrible at visualizing four dimensions. So, we'll move back to 3D space and imagine that we're trying to measure the distance between two points on a 3D grid. This time, however, we'll work with a limitation. Let's say that the two points are confined to the surface of a sphere and for that matter, so are we. In this situation, if we're confined to the surface of the sphere — that is, we can't drill a hole inward or fly up into the air — the only valid distance we can measure is an arc length. This distance here:

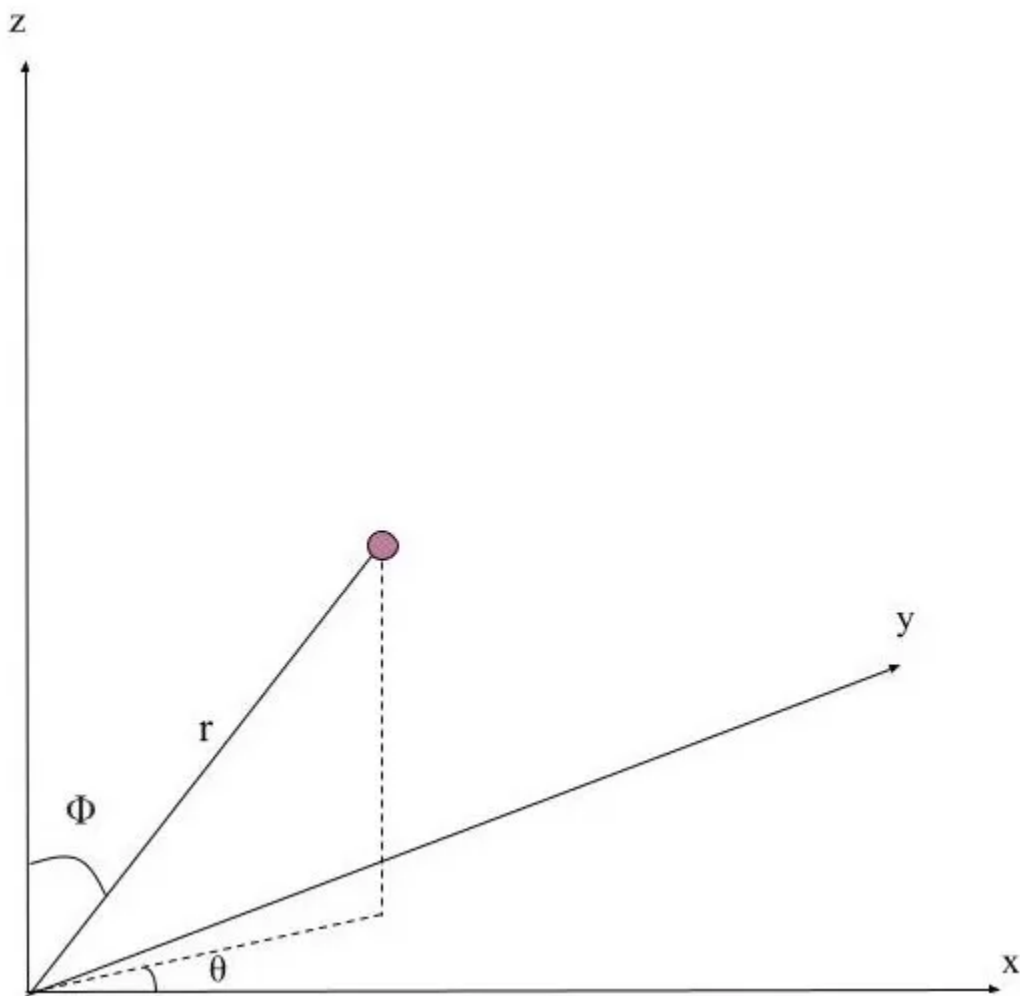


If we're restricted to just the surface of the sphere, the arc length, ds , is equal to the shortest distance between A and B. Now, in such a situation, it turns out that it's incredibly useful to define a new coordinate system. Let's momentarily throw the sphere out. We get back a point on the three-dimensional cartesian system. We can also throw out the conventional x-, y-, and z-cartesian system and work with new generalized coordinates. The first coordinate we'll define is the radial coordinate, r . This represents the distance between the origin and the point. The second coordinate is known as the theta coordinate. It represents the angle between the point's projection on the xy-plane and the x-axis. Finally, we have the last coordinate: ϕ . It's the angle between our point and the y-axis.



The r coordinate, the θ coordinate, and the Φ coordinate.

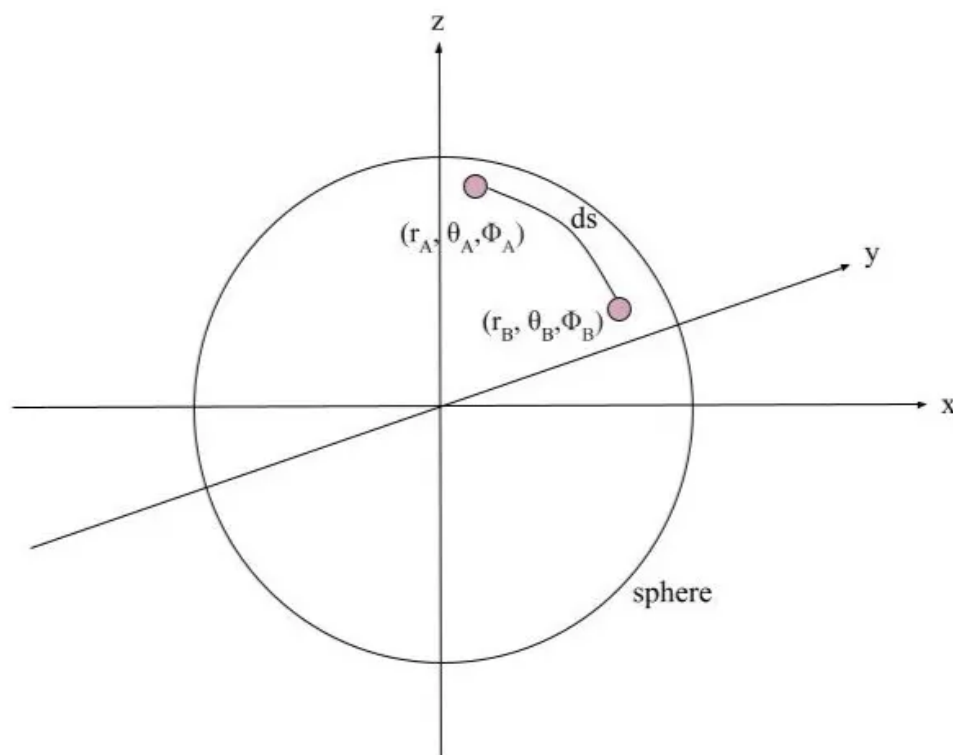
The entire system converges to this:



The spherical polar coordinate system

$$(x, y, z) \rightarrow (r, \theta, \phi)$$

So, we've defined a new coordinate system. We had x, y, z , and now we have r, θ, Φ . This new set of coordinates is known as the spherical polar coordinate system. Now, let's bring back the sphere we were working with.



Since both points A and B lie on the surface of the sphere, we know that their radial components will be the same. $r_a = r_b$. Therefore, the only things we'll need to consider when measuring the distance between these two points is the theta coordinate and the phi coordinate. And of course, this is different for the two points on the sphere. So, we can use this information to find the curved distance, ds , we were trying to work out earlier. Using some clever maths it can be shown that this is

what we're trying to measure:

$$ds^2 = r^2(d\theta^2 + \sin^2 \theta d\phi)$$

This, evidently, is not the same as $dx^2 + dy^2 + dz^2 = ds^2$. That will simply be a straight line — meaningless if we're stuck on a sphere. The interesting question though is that could an object stuck on the surface of a sphere tell that it's on a curved surface without having to look at it in three dimensions. We can apply that question to ourselves too. Could we, without having to escape to a “fifth” dimension, realize whether we're in flat spacetime or curved spacetime.

Well, mathematically, it can certainly be done. We know that gravity causes a curvature in spacetime. We can also if we'd like, calculate *how much* our spacetime has curved — at least in the region of space around bodies with a mass. Now, we, light, planets, the stars, and virtually everything in the cosmos takes the shortest path between two different events in spacetime. However, it's all limited by just how much the spacetime is curved. So, now that we've made it through this, let's move on to the Schwarzschild metric. There are multiple forms that the metric can take. Rewritten, we get something that looks like this:

$$ds^2 = - \left(1 - \frac{2Gm}{r} \right) dt^2 + \frac{dr^2}{\left(1 - \frac{2Gm}{r} \right)} + r^2(d\theta^2 + \sin^2 \theta d\phi)$$

The first version uses units such that $c=1$, while this one does not. This is a common thing to do as well as setting $G=1$. So you're off by a tiny term here. Unconventional but it works.

This equation, you'd notice, also uses the r , θ , and ϕ coordinate system. It's a four-dimensional equation — accounting for four quantities. All of the terms that “scale” the dt^2 and dr^2 are what quantify the curvature of spacetime due to gravity. Specifically, this curvature is credited to the gravity of a spherical object at the center of the coordinate system. Point is that this equation gives us the interval between two events in curved space-time. We achieve this version of the

Schwarzschild metric by accounting for the Schwarzschild radius r_S :

$$r_S = \frac{2Gm}{c^2}$$

The Schwarzschild equation

The Schwarzschild radius, in principle, is the radius to which any body with mass, m , must be squished to achieve a black hole. The Schwarzschild radius for the Earth is **approximately one inch**, meaning that you could squish the entire mass of the Earth into a sphere the size of a basketball and still not have a black hole: light emitted from that mass can still escape the intense gravitational pull.

This tells us that measuring any two events doesn't solely depend on where they are relative to each other. It also depends on where they are with respect to our coordinate system, i.e., our sphere. This works really well for stationary black holes but it's not limited to just that. The Schwarzschild metric is a theoretically sound description of spacetime behavior around a black hole. You'd notice, actually, that the Schwarzschild radius is encoded into our interval equation.

$$\begin{aligned} ds^2 &= -c^2 d\tau^2 \\ &= -\left(1 - \frac{r_S}{r}\right) c^2 dt^2 + \left(1 - \frac{r_S}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \end{aligned}$$

The Schwarzschild metric

What's funny is that Brian Greene, the American theoretician, posted the Schwarzschild metric on the day that the Event Horizon Telescope announced the first picture taken of a black hole. The black hole in question, however, is suspected to be a rotating one. For that, we'd use something called the Kerr metric. It looks a lot more unfriendly:

$$ds^2 = -c^2 d\tau^2$$
$$= -\left(1 - \frac{r_s r}{\Sigma}\right) c^2 dt^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \left(r^2 + a^2 + \frac{r_s r a^2}{\Sigma}\right) \sin^2 \theta d\phi^2 - \frac{2r_s r a \sin^2 \theta}{\Sigma} c dt d\phi$$

The Kerr metric

Anyway, with that, I hope you have a fairly intuitive grasp of what the Schwarzschild metric does and what it means. Feel free to tear me apart in the comments if I've made any errors. And as always, thank you for reading!

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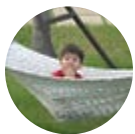


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